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Aerial GCP Pose Estimation using YOLOv8

Overview

This project addresses the problem of automatically localizing Ground Control Points (GCPs) and classifying their marker shapes from aerial drone imagery.

Ground Control Points are surveyed reference markers used in photogrammetry and mapping workflows. Accurate identification of their center coordinates is critical for georeferencing and 3D reconstruction pipelines.

The objective of this assignment was to:

- Predict the center coordinates (x, y) of a GCP marker.
- Classify the marker shape as one of:
 - Cross
 - Square
 - L-Shape

Dataset Analysis

The provided dataset consisted of aerial images organized in a nested directory structure along with annotation data stored in a JSON file.

Dataset Summary

- Total annotations: 996
- Available images after validation: 605
- Number of classes: 3
 - Cross
 - Square
 - L-Shape

Data Quality Challenges

During Exploratory Data Analysis (EDA), the following issues were identified:

- 4 records were missing shape labels.
- 19 records contained invalid or out-of-bound coordinates.
- Images were stored in a deeply nested folder structure.
- Class distribution was imbalanced, with fewer Cross markers compared to other classes.

These issues were addressed during preprocessing to ensure reliable training.

Methodology

Model Selection

YOLOv8s was selected due to its:

- Strong localization capability
- Fast inference speed
- Lightweight architecture
- Ease of deployment

Although the original dataset provided only center-point annotations, YOLO requires bounding boxes for training. Therefore, fixed-size bounding boxes were generated around each annotated center point.

Data Preparation

The preprocessing pipeline included:

1. Loading and validating annotations.
 2. Removing invalid records.
 3. Correcting out-of-bound coordinates.
 4. Creating training and validation splits.
 5. Generating YOLO-compatible annotation files.
 6. Creating a structured dataset for training.
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Training Configuration

Parameter Value

Model YOLOv8s

Epochs 150

Image Size 960

Batch Size 8

Hardware NVIDIA Tesla T4

Results

Validation Metrics

Metric	Score
Precision	0.7564
Recall	0.5933
mAP@50	0.6507
mAP@50-95	0.3988

Training Artifacts

The training_results directory contains:

- Training curves
 - Precision-Recall curves
 - F1 score curves
 - Confusion matrices
 - Training logs
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Inference Strategy

For each detected marker:

1. YOLO predicts a bounding box.
2. The center of the predicted bounding box is used as the estimated GCP location.

3. The predicted class label is mapped to:

- Cross
- Square
- L-Shape

The final predictions are exported in the required JSON format.

Project Structure

Skylark_Assignment/

├── GCP_Pose_Estimation.ipynb

├── README.md

├── predictions.json

├── best_gcp_detector.pt

└── training_results/

Included Files

GCP_Pose_Estimation.ipynb

Contains:

- EDA
- Data preprocessing
- Dataset preparation
- Training pipeline
- Inference workflow

best_gcp_detector.pt

Best trained YOLOv8 model weights.

predictions.json

Predictions generated on the provided test dataset in the required submission format.

training_results/

Contains all training metrics, visualizations, and evaluation outputs.

Reproducing Inference

```
from ultralytics import YOLO
```

```
model = YOLO("best_gcp_detector.pt")
```

```
results = model.predict(  
    source="path/to/image.jpg",  
    conf=0.25  
)
```

Assumptions

- Fixed-size bounding boxes were generated around annotated center points because only keypoint coordinates were provided.
- The center of the predicted bounding box was treated as the estimated GCP center during inference.
- Records with missing labels were excluded from training.

Conclusion

A complete end-to-end pipeline was developed for GCP localization and shape classification. The solution includes data validation, preprocessing, model training, evaluation, and prediction generation, producing outputs in the format required for assessment.